Niyar R Barman

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Education

2021 - 2025 **National Institute of Technology Silchar, India**, *Bachelors of Technology in Electronics and Communication Engineering*

2022 - 2025 **Indian Institute of Technology Madras, India**, *Bachelors of Science in Programming and Data Science*

Experience

Oct 2023 - Present **Research Intern** , *Xu Lab at Carnegie Mellon Univeristy Computational Biology Department*, United States

- Working on deep learning models for Cryo-ET data analysis and exploring the possibilty of implementing a Visual Question Answering (VQA) pipeline tailored for single particles from Cryo-EM.

Jan 2023 - Present **Research Intern** , *Artificial Intelligence Institute of UofSC,*, United States

- CT^2: Analyzed 100K data points with 15 language models to determine negative log curvature and perplexity values, performed a range of statistical tests, and developed task-specific pipelines.
- AGID: Implemented various existing 2D watermarking methods from scratch and devised strategies for adversarial attacks and visual paraphrasing to remove watermarks from images.
- DeHATE: Developed a pipeline for generating hateful images through stable diffusion, integrated DAAM to pinpoint areas impacted by hate words, and applied mosaic blurring to censor these regions. Additionally, generated hateful images using a hate speech dataset for analysis.

Publications

EMNLP 2023 **Counter Turing Test CT^2: AI-Generated Text Detection is Not as Easy as You May Think – Introducing AI Detectability Index (2023)**, *Megha Chakraborty, S. M Towhidul Islam Tonmoy, S M Mehedi Zaman, Krish Sharma,* ***Niyar R Barman****, Chandan Gupta, Shreya Gautam, Tanay Kumar, Vinija Jain, Aman Chadha, Amit P. Sheth, & Amitava Das.*

Honored with the Outstanding Paper Award in the Resources and Evaluation Track

IEEE CICT 2023 **A Transformer-Based Approach to Automate Disease Prediction from Patient Descriptions (2023)**, ***Niyar R Barman****, Krish Sharma & Ranjay Hazra.*

FIRE 2023 **Addressing Hate Speech: ATLANTIS for Efficient Hate Span Detection (2023)**, ***Niyar R Barman***, *Krish Sharma, Yashraj Poddar, Advaitha Vetagiri & Partha Pakray.*

Open Source Contributions

pytorch/examples: Implemented Vision Transformer using only native PyTorch libraries, trained on the CIFAR-10 dataset

Technical Skills

Programming languages: Python, JavaScript, C, C++, Bash

Libraries & Frameworks: PyTorch, TensorFlow, Keras, Scikit, Django, FastAPI, ReactJS, NextJS, React Native

Tools: Git, Docker

Miscellaneous: AWS, LATEX, Linux, Windows, Blender

Key Projects

swiftGPT

- Developed SwiftGPT, a natural language generation model based on the GPT-2 architecture, specifically trained on an artist's songs dataset to emulate their writing style.
- Implemented a custom BigramLanguageModel using PyTorch, incorporating multi-head self-attention and feed-forward layers. Conducted 5000 training iterations with an AdamW optimizer and monitored train and validation loss to fine-tune the model for creative songwriting.

Viper

- Applied Reinforcement Learning and Deep Q Learning principles to develop an AI capable of playing the classic game Snake.
- Implemented the project from scratch using Pygame for the game environment and PyTorch for neural network-based decision-making.

MapSnap

- Developed MapSnap, a web application that leveraged the Segformer model for real-time semantic segmentation of satellite images. The application generated masked outputs that highlighted affected areas
- Successfully fine-tuned the Segformer model using the IARAI landslide4sense dataset, resulting in accurate and efficient identification of landslide-prone areas

DiagnoAI

- Fine-tuned the BERT transformer model to detect 24 common diseases and generate probability scores for the top 3 diseases.
- Curated and labeled a comprehensive dataset of 1200 symptom descriptions associated with 24 diseases, which was used for training and validating the model.

Awards

Aug 2023 **2**nd in HateNorm'23, a component of HASOC 2023, by excelling in Task 3, which involved the identification of hateful spans within sentences already categorized as hateful.

Apr 2023 **1**st in Neurathon'23, the annual machine learning hackathon organized by the ML Club at NIT Silchar.

Jul 2022 **4**th in Un-Flood Assam'22 hackathon organized by the Ministry of Electronics & Information Technology.

Positions of Responsibility

Dec 2023 **EMNLP Presenter**: Presented Counter Turing Test CT^2 (Winner of Outstanding Paper award in the Resources and Evaluation Track) at EMNLP Main Poster Presentation 2023.

Jun 2023 - Present **Teaching Assistant/Moderator**: Conducted introductory classes on Python and Machine Learning for freshmen and sophomores as part of the Machine Learning Club at NIT Silchar.